Neural Decoding of Natural versus Synthetic Visual Inputs with PCA and Supervised Learning

Richard Li
University of Washington
Seattle, WA 98195
mcl618@uw.edu

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1 Introduction

Understanding how visual stimuli are transformed into distributed cortical responses remains a central problem in computational neuroscience. Recent model-based studies suggest that low-dimensional representations can capture meaningful neural structure and support decoding [Yamins and DiCarlo, 2016, Wen et al., 2018, Guclu and van Gerven, 2015]. This project asks whether natural-versus-synthetic stimulus information can be recovered from reduced fMRI response space, and whether stimulus-side image features can be mapped back to neural principal components.

To address this, the analysis uses the Natural Scenes Dataset (NSD) and NSD-Synthetic extension as paired image-response sources [Allen et al., 2022, Gifford et al., 2026]. The workflow combines exploratory dimensionality analysis (PCA/FA), supervised decoding, and feature-to-neural encoding. The objective is not only high classification performance, but also a clearer view of when simple linear structure is sufficient and when additional model capacity is needed [Haxby et al., 2001, Kamitani and Tong, 2005, Norman et al., 2006].

2 Data Description

The study uses trial-level NSD/NSD-Synthetic fMRI responses paired with image stimuli. Responses are masked by a resampled brain volume (`resampled_mask.npy`, shape $81 \times 104 \times 83$), yielding $V = 429{,}603$ voxel features per trial. Core matrices are `X_syn.npy` $(744, 429603)$ and natural-session matrices `X_nat_s1.npy`, `X_nat_s2.npy`, `X_nat_s3.npy` (each $(750, 429603)$).

Binary labels are defined as natural $= 0$ and synthetic $= 1$. Decoder training uses synthetic plus natural session-1 only, with an 80/20 split into `X_train.npy` $(1195, 429603)$ and `X_test.npy` $(299, 429603)$ (train counts $\{0 : 586, 1 : 609\}$; test counts $\{0 : 164, 1 : 135\}$). A pooled evaluation view additionally merges natural sessions 2–3 with synthetic responses in shared PCA space, producing `X_all_pca.npy` $(2244, 100)$ and `y_all.npy` with class counts $\{0 : 1500, 1 : 744\}$; this imbalance is considered in interpretation.

3 Methods

The pipeline follows four phases (`01_data.ipynb`–`04_cnn.ipynb`): data assembly, dimensionality reduction, decoding, and encoding. Trial-wise masked responses are saved to `outputs/processed`, then shuffled (NumPy seed 42) and split 80/20 for decoder training and testing.

fMRI responses are standardized using training statistics only,

$$\tilde{\mathbf{x}} = \frac{\mathbf{x} - \boldsymbol{\mu}_{\text{train}}}{\boldsymbol{\sigma}_{\text{train}}},$$

and reduced with SVD-based PCA to $k = 100$ components (from 429,603 voxel features), yielding `PCs.npy`, `X_train_pca.npy`, and `X_test_pca.npy`. Decoder models compare logistic regression (single linear layer, cross-entropy) against a one-hidden-layer MLP (width 100), evaluated with accuracy, confusion matrices, and PC-space error localization.

For encoding, image features are extracted from a pretrained ResNet-18 penultimate layer (512-D features). Synthetic image features are aligned with synthetic fMRI PCA responses, and regression models map image features to neural PCs:

$$\hat{\mathbf{u}} = f_{\phi}(\mathbf{z}),$$

where $\mathbf{z} \in \mathbb{R}^{512}$ and $\hat{\mathbf{u}} \in \mathbb{R}^{100}$. Linear regression, ridge regression (Adam with weight decay), and an MLP regressor are compared using test MSE and per-PC R^2.

4 Results

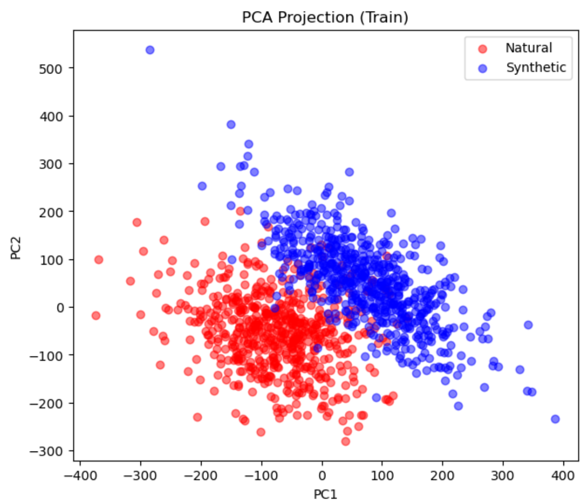
PCA Reveals Partial Separation Between Natural and Synthetic fMRI Responses Without Dominant Latent Confounds

PCA is used to assess whether condition structure is visible before supervised decoding. Figure 1a and Figure 1b show partial natural/synthetic separation with overlap near the center of PC space.

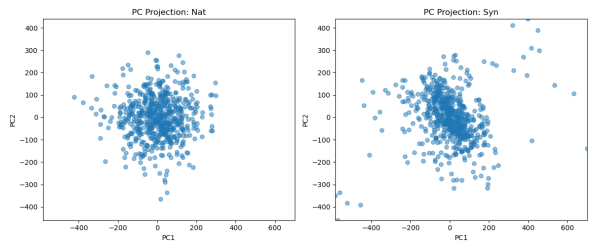
The PCA projections indicate that class structure is not random in the leading neural dimensions. Natural and synthetic trials show partial geometric separation with overlap concentrated near the center of the plane. The condition-resolved view also suggests a dispersion difference, where one class forms a tighter cluster and the other is more diffuse, consistent with different levels of response consistency across trials.

Figure 2 shows stronger class mixing in FA space than in PCA space, indicating that dominant shared latent factors capture common variance rather than class-discriminative structure. In other words, FA emphasizes low-rank structure shared across conditions, while PCA preserves directions that are more useful for discrimination.

Together, these views support the decoder stage: condition signal is present, but concentrated in partially overlapping low-dimensional geometry rather than cleanly separable clusters.



(a) Overall PCA behavior on the training responses.



(b) Condition-wise PCA structure for natural and synthetic trials.

Figure 1: PCA-based exploratory views of the masked fMRI response space.

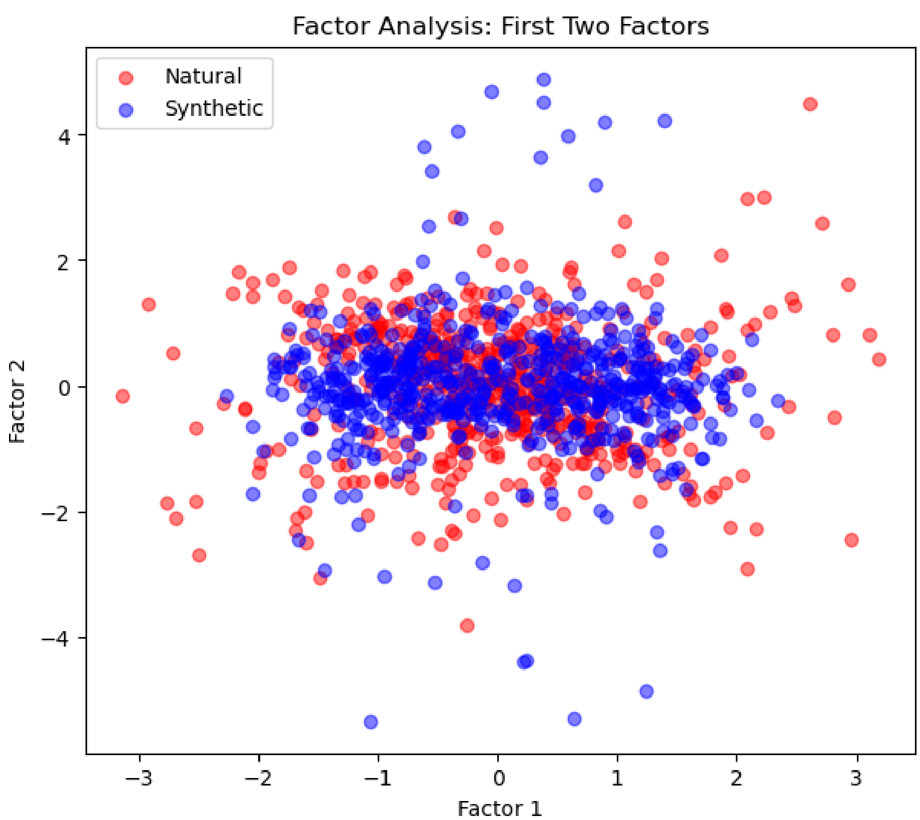


Figure 2: Factor-analysis visualization of shared latent structure in the masked fMRI responses.

Decoding with PCA Geometry Favors Linear Models

Decoder performance is evaluated through PC-space error localization (Figure 3), confusion matrices (Figure 4), and training curves (Figure 5). This combination distinguishes geometric failure modes from aggregate class outcomes and optimization dynamics.

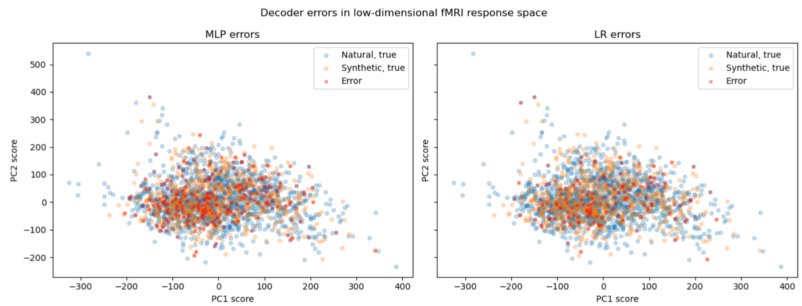


Figure 3: Error localization for MLP and logistic regression in the first two principal-component dimensions.

Figure 3 shows that most errors cluster in the overlap region of PC1–PC2, consistent with intrinsic class mixing rather than isolated outlier failures. The logistic-regression panel exhibits fewer overlap-region errors than MLP, suggesting that a linear boundary is better aligned with the dominant discriminative axes in this reduced space.

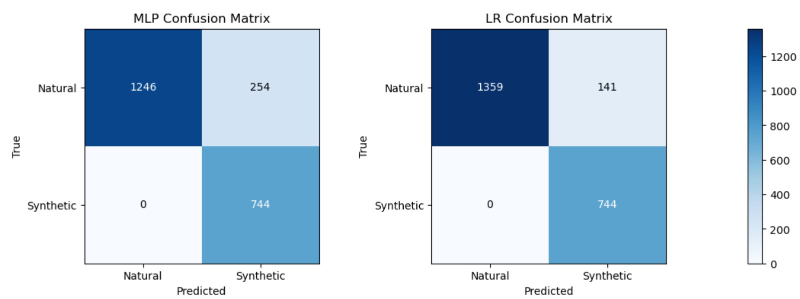


Figure 4: Confusion matrix comparison for MLP and logistic regression classifiers.

Figure 4 quantifies the gap: logistic regression yields 2103/2244 correct predictions versus 1990/2244 for MLP. Both recover synthetic trials strongly (0 synthetic-to-natural errors), and the performance difference is driven by natural-to-synthetic overprediction (141 for logistic regression vs 254 for MLP). This asymmetric error pattern indicates that additional nonlinearity does not improve discrimination in the current representation and can increase false-positive synthetic calls.

Figure 5 shows smoother MLP training loss, but this does not translate to better classification. Overall, decoder results indicate that linear boundaries are sufficient for most recoverable condition structure in the current PCA space, with remaining errors dominated by intrinsic overlap rather than model under-capacity.

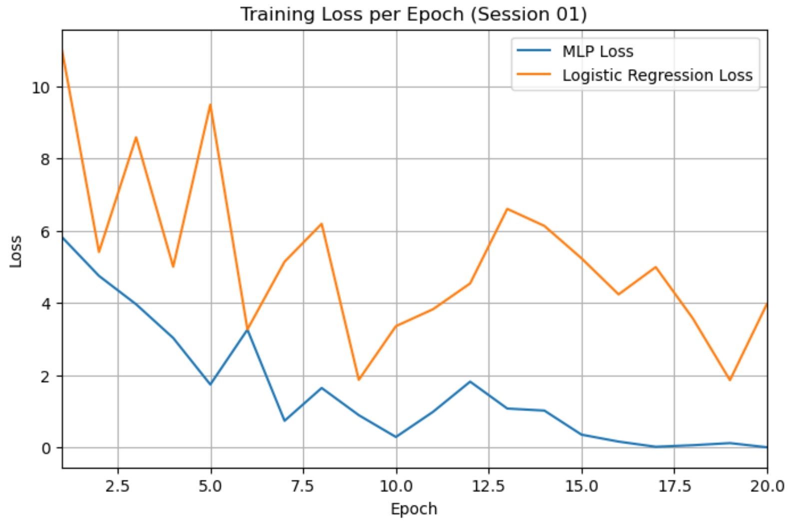


Figure 5: Subject-level comparison between MLP and logistic regression (subject 01).

CNN Feature Geometry Shows Partial Alignment With Neural Structure

Encoder analysis tests whether compact CNN features preserve structure predictive of neural PCs and whether this feature space supports stable mapping back to neural principal components.

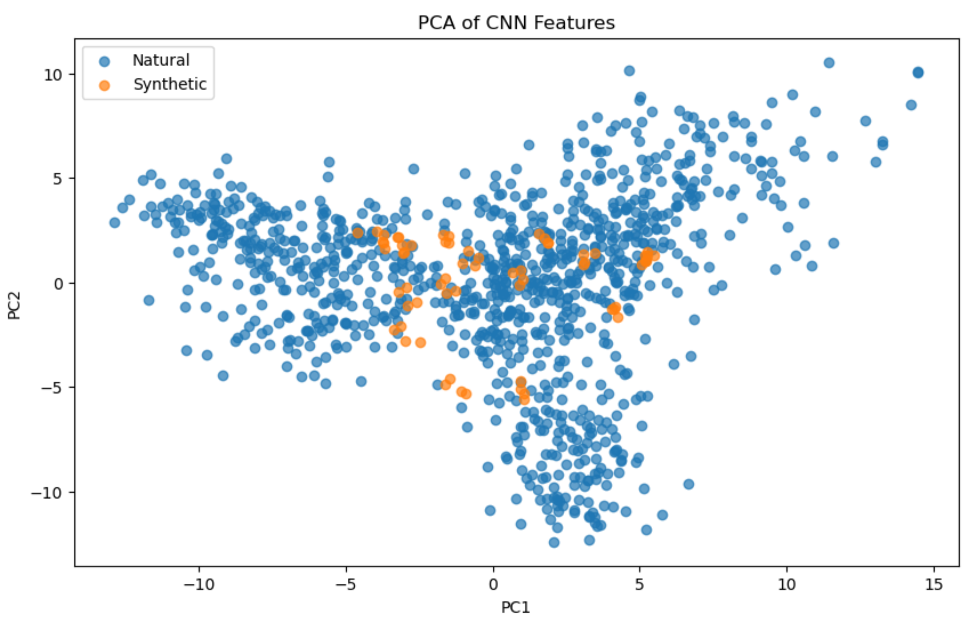


Figure 6: Low-dimensional structure of CNN-derived image features.

In Figure 6, synthetic samples occupy a compact region within a broader natural-image cloud, suggesting partial overlap in feature space but reduced coverage of natural-scene variability. This geometry implies that shared low-level feature axes are present, but the synthetic set does not span the full diversity of natural-scene structure.

Ridge Regression Is the Most Reliable Encoder Across Neural Components

The model comparison is summarized in Figure 7 (predicted-versus-actual values for the first three PCs) and Figure 8 (per-component R^2 across all 100 neural PCs).

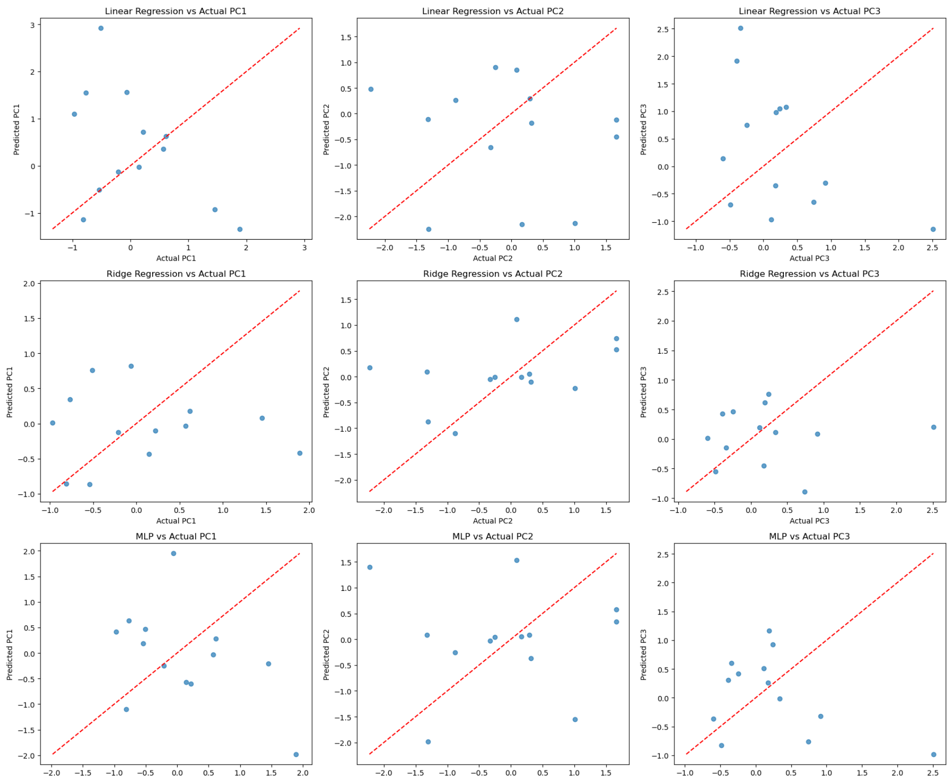


Figure 7: Predicted-versus-actual comparison for linear, ridge, and MLP encoders on the leading neural principal components.

Figure 7 and Figure 8 both favor ridge: predictions are closer to diagonal for leading PCs and R^2 is more stable across components. Mean R^2 values are -0.8506 (linear), 0.2276 (ridge), and -0.1505 (MLP), confirming ridge as the best encoder under limited paired samples.

Negative mean R^2 for linear regression and MLP indicates performance below a mean-predictor baseline on average across components, while ridge remains positive and comparatively stable. This supports ridge regularization as the strongest bias-variance tradeoff in the current small-sample encoder regime.

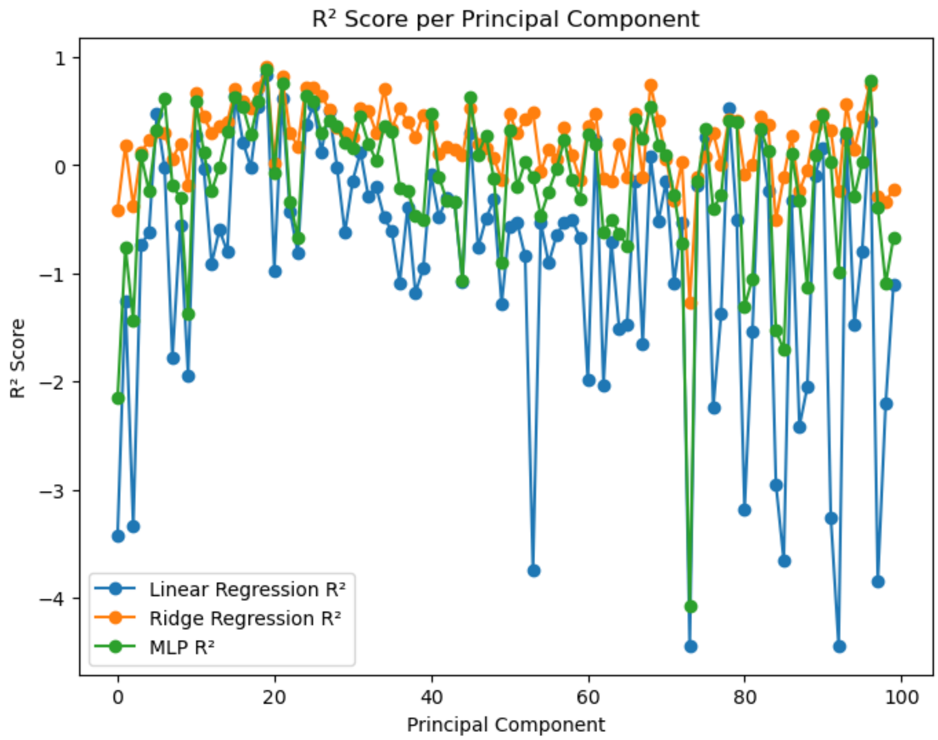


Figure 8: Per-component R^2 comparison across all 100 neural principal components.

5 Discussion

Results from both decoder and encoder analyses indicate that meaningful condition information is preserved in reduced neural space, and that regularized linear models are strong baselines in this regime. Logistic regression matched the low-dimensional geometry of the decoder task, while ridge regression provided the most stable encoding behavior across neural components.

The findings should be interpreted with key constraints in mind: pooled labels are imbalanced (natural $n = 1500$ vs synthetic $n = 744$), development is effectively single-subject/session-centered, and model selection was limited by compute and memory budgets. Future work should prioritize balanced multi-subject protocols and higher-capacity mappings (e.g., transformer-based feature extractors and cross-attention decoders) to test which image-to-neural relationships generalize beyond this within-subject baseline.

6 Conclusion

An end-to-end pipeline was developed to study neural decoding and encoding of natural versus synthetic visual stimuli using NSD/NSD-Synthetic responses. Across exploratory analyses, decoder evaluation, and encoder modeling, the results converged on a consistent finding: useful discriminative structure is present in low-dimensional neural representations, and simple regularized models capture that structure more reliably than higher-capacity alternatives in the current data regime. In decoding, logistic regression aligned well with PCA geometry and outperformed the MLP in pooled confusion-matrix behavior, indicating that most remaining errors are attributable to intrinsic class overlap rather than insufficient nonlinear capacity. In encoding, ridge regression achieved the strongest and most stable reconstruction profile across principal components, including the highest mean R^2, supporting regularization as a key requirement for small-sample image-to-neural mapping.

At the same time, conclusions are bounded by class imbalance, single-subject development, and computational constraints that limited model scale and search depth. The framework therefore serves as a reproducible baseline rather than a final account of biological visual computation. Future work should extend to balanced multi-subject protocols and higher-capacity transformer-based mappings from raw or high-dimensional image features to neural response space, with stronger uncertainty and stability analysis to test which representational relationships generalize robustly across subjects and conditions.

7 Acknowledgments

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